



RwaCow — Deployment Guide (Free Hosting)

Deploy the system for free using **Railway** (backend + MySQL) and **Vercel** (frontend).

Result: a public URL you can open from any device to test the system online.

Overview

Part	Host	Cost
Frontend (React)	Vercel	Free
Backend (Node/Express)	Railway	Free trial credit
Database (MySQL)	Railway	Free trial credit

Prerequisites

- A **GitHub account** (github.com)
- A **Railway account** (railway.app) — sign up with GitHub
- A **Vercel account** (vercel.com) — sign up with GitHub
- **Git** installed on your computer (git-scm.com)

Step 1 — Push Your Code to GitHub

From the project root (`E:\Cow-management`):

```
git init
git add .
git commit -m "Initial commit - RwaCow system"
```

Then create a new **empty** repository on GitHub (e.g. `rwacow`), and:

```
git remote add origin https://github.com/YOUR_USERNAME/rwacow.git
git branch -M main
git push -u origin main
```

> The `.gitignore` already excludes `node_modules` , `.env` , and the local database — good.

Step 2 — Deploy the Database + Backend on Railway

2a. Create the MySQL database

- Go to railway.app → **New Project**
- Click **Provision MySQL** (or "Add a Database" → MySQL)
- Railway creates a MySQL instance and gives you connection details

2b. Deploy the backend

- In the same project, click **New** → **GitHub Repo** → select your `rwacow` repo
- Railway detects Node.js. Set the **Root Directory** to `backend`
- Go to the backend service → **Variables** tab and add:

```
DB_ENGINE = mysql
MYSQL_URL = ${MySQL.MYSQL_URL}
JWT_SECRET = <a-long-random-secret-string>
JWT_EXPIRES_IN = 7d
NODE_ENV = production
FRONTEND_URL = https://your-frontend.vercel.app
```

> `MYSQL_URL` references the MySQL service Railway created. Use Railway's
> variable reference feature, or copy the connection URL directly.

- Under **Settings** → **Deploy**, set the **Start Command** to:

...

```
node src/index.js
```

...

2c. Create the tables + seed data (run once)

In the backend service, open the **Railway shell** (or use a one-off command) and run:

```
npm run deploy:setup
```

This creates all tables and adds the test accounts.

2d. Get your backend URL

Railway gives the backend a public URL like:

```
https://rwacow-backend-production.up.railway.app
```

Your API base is that URL + `/api`.

Step 3 — Deploy the Frontend on Vercel

- Go to vercel.com → **Add New** → **Project**
- Import your `rwacow` GitHub repo

- Set **Root Directory** to `frontend`
- Vercel auto-detects Vite. Confirm:
- Build command: `npm run build`
- Output directory: `dist`
- Add an **Environment Variable**:

```
VITE_API_URL = https://your-backend.up.railway.app/api
```

(use your actual Railway backend URL from Step 2d)

- Click **Deploy**

Vercel gives you a public URL like:

```
https://rwacow.vercel.app
```

Step 4 — Connect Frontend ↔ Backend

- Copy your **Vercel frontend URL**
- Go back to Railway → backend → **Variables**
- Set `FRONTEND_URL` to your Vercel URL (e.g. `https://rwacow.vercel.app`)
- Railway redeploys automatically

This lets the backend accept requests from your hosted frontend (CORS).

Step 5 — Test It

- Open your Vercel URL in any browser (even on your phone)
- Log in with a seeded account:
- `admin@rwacow.rw` / `admin123`
- If the first request is slow, that's the free-tier backend waking up — it's normal.

Environment Variables Summary

Backend (Railway)

Variable	Value
<code>DB_ENGINE</code>	<code>mysql</code>
<code>MYSQL_URL</code>	(Railway MySQL connection URL)
<code>JWT_SECRET</code>	a long random string
<code>JWT_EXPIRES_IN</code>	<code>7d</code>

Variable	Value
<code>NODE_ENV</code>	<code>production</code>
<code>FRONTEND_URL</code>	your Vercel URL

Frontend (Vercel)

Variable	Value
<code>VITE_API_URL</code>	your Railway backend URL + <code>/api</code>

Alternative: Render (instead of Railway)

If you prefer **Render** for the backend:

- [render.com](#) → **New** → **Web Service** → connect GitHub
- Root directory: `backend` , Build: `npm install` , Start: `node src/index.js`
- Add the same environment variables
- For MySQL, use **Railway MySQL**, **Aiven**, or **Clever Cloud** and put the URL in `MYSQL_URL`

> Note: Render's free tier sleeps after 15 minutes of inactivity and takes

> ~30 seconds to wake on the first request.

Troubleshooting

Problem	Fix
Frontend shows "Demo Mode"	Backend URL wrong or backend asleep. Check <code>VITE_API_URL</code> and wait for wake-up
CORS error in browser console	<code>FRONTEND_URL</code> on the backend doesn't match your Vercel URL
Login fails online	Run <code>npm run deploy:setup</code> on Railway to seed accounts
Database connection error	Check <code>MYSQL_URL</code> is set correctly on the backend
Build fails on Vercel	Ensure Root Directory is <code>frontend</code>

Updating the Deployed App

Whenever you push changes to GitHub:

```
git add .
git commit -m "your change"
git push
```

Both Railway and Vercel **auto-redeploy** from the `main` branch. No manual steps needed.

